

Keaton Kraiger

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Education

- Pennsylvania State University**, PhD in Computer Science Engineering Jan. 2021 – Expected 2026
- Research topics: Computer vision, multimodal learning, and learning human dynamics from video
 - Advisors: Dr. Yanxi Liu and Dr. Robert T. Collins
- Portland State University**, BS in Computer Science (Magna Cum Laude) Aug. 2016 – June 2020
- Advisor: Dr. Dan Hammerstrom

Research Interests

I am broadly interested in machine learning and its applications in computer vision, graphics, and human motion understanding. My research focuses on multimodal learning in video, sensor-based, and Mocap domains, applying it to dynamics from human motion.

Research Experience

- Graduate Research Assistant**, Pennsylvania State University – State College, PA Jan. 2021 – Present
Laboratory for Perception, Action, and Cognition (LPAC)
- Advisors: Dr. Yanxi Liu and Dr. Robert Collins
 - Designed and deployed a multimodal data collection system integrating ROS2-based RGB-D camera arrays, Meta Project Aria smart glasses, and insole pressure sensors
 - Estimated human stability from multimodal data (video, motion capture, and foot pressure)
 - Collected, cleaned, and released motion capture, video, and foot pressure dataset of over 60k frames
 - Performed multimodal action recognition and established classification dataset and benchmark
 - Developed class-agnostic object detection method and its downstream application in enhancing captions
- Undergraduate Research Assistant**, Portland State University – Portland, OR Dec. 2018 – Jan. 2020
Biologically-Inspired Computing Lab (BICL)
- Advisor: Dr. Dan Hammerstrom
 - Develop biologically-inspired algorithms to perform object detection
 - Implemented brain-inspired algorithms to detect objects with position and scale invariance
 - Process image datasets with grid cell model to aid in object detection and image classification
 - Compare different image classifier performances, specifically convolutional neural networks and capsule networks when integrated with the grid cell model
- Undergraduate Mentee**, Portland State University – Portland, OR Nov. 2018 – May 2019
- One of four students selected to participate in the Undergraduate Research & Mentoring Program
 - Paired with faculty mentor to conduct funded research during the winter and spring term
 - Attended program workshops on developing abstracts, research proposals, research questions, research methods, and means of communicating research findings

Publications

- FootFormer: Estimating Stability from Visual Input** Nov. 2025
Keaton Kraiger, Jingjing Li, Skanda Bharadwaj, Jesse Scott, Robert T. Collins, Yanxi Liu.
BMVC 2025
- Novel 3D Scene Understanding Applications From Recurrence in a Single Image** Aug. 2023
Shimian Zhang, Skanda Bhara, **Keaton Kraiger**, Yashasvi Asthana, Hong Zhang, Robert T. Collins, Yanxi Liu

Work Experience

Head Teaching Assistant, Pennsylvania State University – State College, PA Aug. 2021 – Dec. 2023

- Head TA for Vision and Language, Computer Vision I & II, and Machine Learning
- Designed and implemented course projects focused on vision, deep learning, and reinforcement learning

Technical Course Specialist, Portland State University – Portland, OR Sep. 2017 – June 2020

- Led homework recitation sessions on undergraduate programming assignments and algorithm development
- Provide feedback and grades on programming and written assignments

Presentations and Posters

Learn From Human Eyes: Zero-Shot Recurring Pattern Detection on a Multi-Perception Benchmark May 2025

Shimian Zhang, **Keaton Kraiger**, Skanda Bharadwaj, Robert Collins, Yanxi Liu
Symmetry 2025 Abstract Track | Slides
Best Oral Presentation Award

Vision to Dynamics February 2025

Keaton Kraiger, Skanda Bharadwaj, Yanxi Liu, Robert T. Collins
NYC Vision Day 2025 | Poster

Vision to Dynamics April 2024

Keaton Kraiger, Yuan Gao, Jeff Koumba, Yanxi Liu, Robert T. Collins
NYC Vision Day 2024 | Poster

Humanoid Robots (Real and Simulated) Feb. 2024

Yuan Gao, Jeff Koumba, **Keaton Kraiger**
Pittsburgh Robotics Network Discovery Day 2023 | Poster

The Applications of Grid Cells in Computer Vision, April 2019

Keaton Kraiger, Dan Hammerstrom
Portland State University Student Research Symposium | Poster

Additional Experience And Awards

Undergraduate Research & Mentoring Program (URMP)

- Selected to participate in the URMP at Portland State, receiving funding to conduct research with a faculty mentor and receive training on conducting research

Outstanding Teaching Assistant Award

- Recognized by the CSE college for Vision and Language Spring 2023 TA

Conference Reviewer

- Served as a reviewer for multiple years of WACV, ECCV, and CVPR